DS&AI Lab Project [Term Jan 2026]

Action-Oriented Indoor Navigation Assistance for the Visually Impaired

MILESTONE-1: Problem Definition & Literature Review

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1. Introduction and Problem Definition

Indoor navigation remains a significant challenge for visually impaired individuals. Unlike outdoor environments, indoor spaces lack reliable global positioning systems and often contain dense, dynamic obstacles such as furniture, people, and temporary obstructions. As a result, visually impaired users must rely heavily on assistive technologies to safely move within homes, offices, or public buildings.

Most existing assistive systems focus on object awareness, where detected objects are announced through audio feedback. While such systems inform the user about the presence of obstacles, they do not provide guidance on how the user should move in response. For example, an alert such as “chair detected” does not indicate whether the chair lies directly in the walking path, how imminent the risk of collision is, or whether the user should stop or change direction.

This ambiguity often leads to hesitation, frequent stops, increased cognitive load, and in some cases, collisions.

Problem Statement

How can real-time visual perception be converted into clear, actionable movement guidance that enables safe and continuous indoor walking for visually impaired users, without relying on indoor maps or localization systems?

Rather than focusing on identifying what objects are present, this project focuses on determining what action the user should take at each moment.

2. Scope and Assumptions

In-Scope	Out-of-Scope
Indoor environments such as homes and offices	Full indoor localization or position tracking
Near-field navigation within a range of 0–3 meters	Predefined indoor floor maps
Real-time operation using a monocular RGB camera	Simultaneous Localization and Mapping (SLAM)-based mapping and localization
Obstacle collision avoidance and blocked-path handling	Infrastructure-based solutions

Directional movement guidance (e.g., “move left”, “move right”, “stop”)	Long-range route planning or room-to-room navigation
Rule-based, interpretable decision-making	Autonomous navigation without human control
Offline execution without cloud dependency	

Table 2.1: Scopes of Project Aspects

This project is deliberately scoped to address real-time, local indoor navigation assistance rather than full indoor localization or global route planning. The system operates in indoor environments using a monocular RGB camera to provide immediate, action-oriented movement guidance aimed at preventing collisions and enabling continuous walking. To ensure robustness, and interpretability, the design avoids reliance on indoor maps, localization frameworks, or external infrastructure. By focusing on near-field decision-making and rule-based reasoning, the project prioritizes safety-critical assistance over long-range navigation, aligning the system’s capabilities with practical constraints and measurable user benefits.

3. Motivation and Practical Relevance

Indoor mobility is a frequent and high-risk activity for visually impaired individuals. Most collisions and navigation difficulties occur at close range, where immediate and correct movement decisions are required. Delayed or unclear feedback can result in abrupt stops, loss of confidence, or physical injury.

Common scenarios where this problem arises include:

- Walking through corridors
- Navigating around furniture in homes
- Moving within office spaces
- Entering or exiting rooms

Improving action-oriented guidance in these settings can significantly enhance user safety, confidence, and independence.

4. Existing Approaches

Research on assistive navigation for visually impaired users has explored a range of vision-based and sensor-based solutions. These approaches can be broadly categorized into object detection–based systems, proximity warning systems, and localization-driven indoor navigation frameworks.

4.1. Object Detection–Based Assistive Systems

A large body of assistive navigation research employs real-time object detection models such as YOLO, SSD, or Faster R-CNN to identify common indoor obstacles including chairs, tables, doors, and people. These systems typically use a monocular camera mounted on a wearable device or handheld platform and provide audio feedback by announcing detected object categories. ***Eg: OrCam MyEye***

Limitation:

While these systems successfully identify objects, they do not reason about the spatial relevance of detections with respect to the user's walking path. An object announcement alone does not indicate whether the obstacle lies directly ahead, whether it poses an immediate collision risk, or what corrective action the user should take.

4.2. Distance and Proximity Warning Systems

To improve upon object-only feedback, several systems incorporate depth estimation using stereo cameras, RGB-D sensors, or monocular depth models. These systems estimate the distance to detected obstacles and convey proximity information through graded audio signals, sound intensity modulation, or vibration feedback. ***Eg: Sonic Eye Canes***

Limitation:

Although proximity information is provided, the interpretation of these cues remains the responsibility of the user. Distance or intensity-based feedback does not explicitly indicate how the user should move to avoid an obstacle.

4.3. Indoor Navigation with Localization

More advanced indoor navigation systems aim to provide full route guidance by estimating the user's location within a building. These approaches often rely on SLAM, beacon-based localization (BLE, UWB), fiducial markers, or pre-mapped indoor environments. ***Eg: BLE/UWB positioning, SLAM***

Limitation:

Localization-based solutions typically require prior mapping, external infrastructure, or significant computational resources. These requirements limit scalability and deployment flexibility, particularly in everyday indoor environments such as homes or offices.

4.4. Baseline System

The baseline considered in this project reflects the common state-of-practice in many assistive navigation systems. It consists of:

- Real-time object detection
- Distance estimation using depth cues
- Audio feedback limited to object category announcements

The baseline system does not perform any reasoning about walking path obstruction, collision risk, or movement direction. It therefore serves as a suitable reference point for evaluating whether translating perception into explicit, action-oriented navigation guidance provides measurable benefits in safety and usability.

4.5. Evaluation Practices in Prior Research

Prior research on vision-based assistive navigation systems typically evaluates system performance using both perception-level metrics and navigation-level metrics.

i. Perception-level metrics commonly include object detection accuracy and depth estimation error. However, many assistive navigation studies also evaluate user-centric metrics that directly reflect navigation safety and usability.

ii. These navigation-related metrics often include navigation completion time, number of collisions with obstacles, and the number of corrective stops or hesitations made by the user while navigating an environment. Some studies additionally examine the delay between system feedback and user response, particularly when auditory or haptic guidance (touch-based) is provided.

While perception accuracy is important for reliable obstacle detection, navigation-level metrics provide a more direct measure of whether the system effectively supports safe, continuous mobility for the visually impaired users. The evaluation methodology adopted in this project therefore focuses primarily on navigation safety, user responsiveness, and continuity of movement rather than perception accuracy alone.

5. Proposed Direction and Contribution

The review of existing assistive navigation systems reveals a common limitation across object detection–based, proximity warning, and localization-driven approaches. While current systems are effective at detecting obstacles or estimating distance, they largely provide descriptive or warning-based feedback rather than explicit movement guidance.

Object detection systems announce what objects are present, and proximity-based systems indicate how close obstacles are, but neither directly communicates what action the user should take. Localization-based indoor navigation systems attempt to address navigation more comprehensively, but rely on prior maps, infrastructure, or computationally heavy pipelines that reduce robustness and practicality in everyday indoor environments.

This highlights a clear gap: **the lack of real-time**, **action-oriented reasoning** that translates visual perception into immediate, **interpretable movement instructions** for safe indoor walking. This gap motivates the proposed work, which focuses on near-field decision-making rather than full indoor localization or mapping.

6. Proposed Approach and Technical Grounding

6.1. Proposed system's components

The proposed system is designed as a real-time, action-oriented indoor navigation assistant that operates without indoor maps, localization, or external infrastructure. The system focuses on converting visual perception into explicit movement commands that support safe and continuous walking.
At a high level, the system integrates the following components:

- **Monocular RGB Camera Input:**
  A single RGB camera provides real-time visual input, ensuring low hardware complexity for the sake of a proof of concept.
- **Obstacle Detection Module**
  A real-time object detection model based on modern architectures (e.g., YOLO variants) is used to identify common indoor obstacles such as furniture and people. The model will be trained or fine-tuned using **publicly available indoor datasets**, and if required, supplemented with custom data collected in controlled indoor environments to better reflect the evaluation setting.
- **Monocular Depth Estimation**
  A monocular depth estimation model is employed to infer relative distance to detected obstacles within a near-field range (0-3 meters). The depth model will leverage publicly available indoor depth datasets and pretrained weights, with additional fine-tuning or validation on self-collected indoor data if necessary.

  While monocular depth estimation enables distance inference from a single RGB camera, it has several known limitations. Depth predictions derived from monocular imagery may exhibit scale ambiguity (meaning that the absolute scale of the scene cannot always be determined precisely.) Performance may

also be influenced by lighting conditions, reflective surfaces, and regions with limited visual texture.

Because depth estimates may contain uncertainty, the proposed system will rely on conservative distance thresholds and combine depth information with spatial reasoning/ positioning (left/ centre/ right) to reduce the risk of incorrect navigation guidance.

- **Rule-Based Spatial Reasoning**
  The rule-based spatial reasoning module translates perception outputs into navigation instructions by analysing the spatial relationship between detected obstacles and the user's forward walking path.
  To determine movement guidance, the camera frame is divided into three primary regions representing potential movement directions:

Region	Interpretation
Left zone	Potential safe deviation direction
Central walking zone	User's forward walking path
Right zone	Alternative deviation direction

The central walking zone represents the region most closely aligned with the user's forward movement. Obstacles detected within this region are therefore considered potential obstructions in the user's path. Obstacle relevance is determined by examining whether detected object bounding boxes spatially overlap with this central region.

Depth estimation is used to estimate the approximate distance of detected obstacles from the user. The system defines a near-field safety threshold of approximately 1.5 meters for obstacles located within the central walking zone.

This threshold is selected based on typical human walking dynamics and expected response time to auditory instructions. The average human walking speed is approximately 1.2 to 1.4 meters per second. In addition to human reaction time to audio feedback (in this case: interpreting spoken navigation instruction, specifically, is approximately 1 second), the system must also account for processing latency associated with perception, reasoning, and text-to-speech generation. When these delays are considered together, a user may travel roughly 1.5 meters between the moment an obstacle is detected and the initiation of a corrective movement.

Setting the obstruction threshold near this distance allows the system to issue navigation instructions early enough to reduce the risk of collision while maintaining continuous walking.

Navigation decisions are then generated using interpretable rule-based logic. For example:

- If the central walking zone is clear -> continue forward movement
- If an obstacle occupies the central zone and the left region is clear -> instruct “move left”
- If an obstacle occupies the central zone and the right region is clear -> instruct “move right”
- If obstacles occupy all regions or uncertainty exists -> instruct “stop”

This rule-based approach prioritizes interpretability and safety while enabling the system to convert perception outputs into actionable movement guidance.

- **Action-Oriented Audio Feedback**
  Based on the reasoning output, the system generates concise textual navigation commands such as **“move left”**, **“move right”**, or **“stop”**. Safety-critical instructions are prioritized to ensure timely user response and prevent collisions. These prioritized instructions are then used to generate audio output that is conveyed to the user via audio output devices.

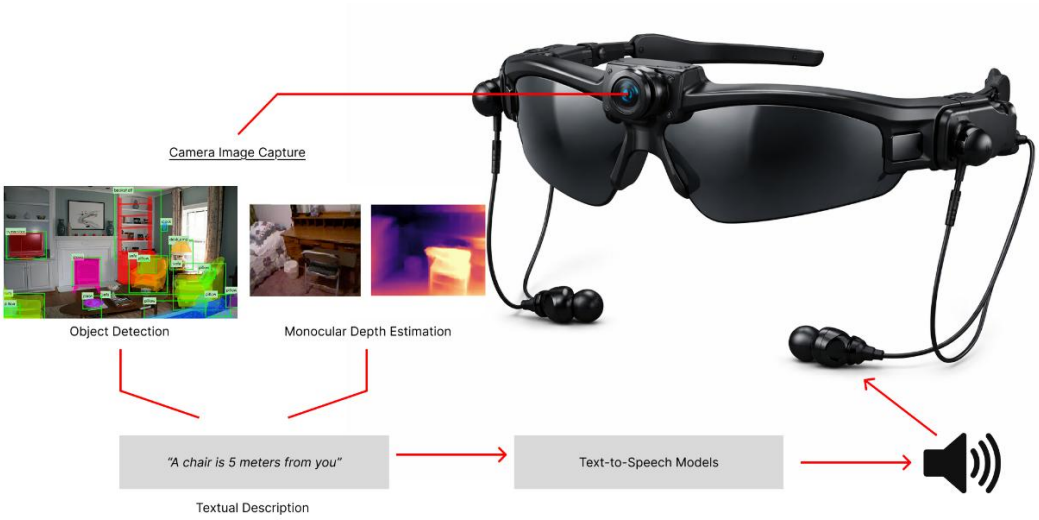
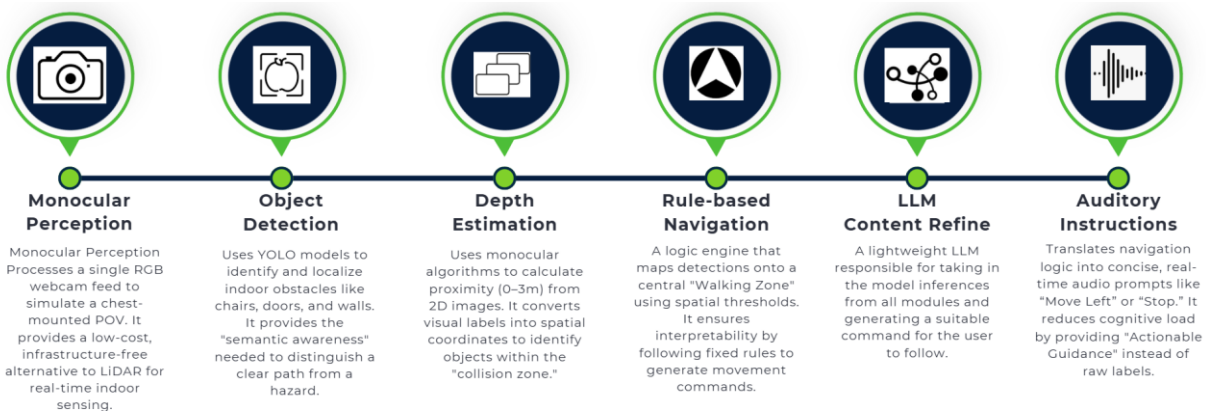


Fig 6.1: Basic ideation of the Project

- **System Reliability and Failure Handling**
  Since the system is intended to support safety- critical navigation, conservative behaviour is adopted in situations where perception results are uncertain. If object detection confidence is low, depth estimates appear inconsistent, or the spatial reasoning module receives conflicting signals, the system will prioritise safety by issuing a conservative instruction such as “stop.”

  This fallback behaviour ensures that uncertain perception outputs do not result in unsafe movement guidance.



  The pipeline operates continuously on incoming camera frames, generating navigation instructions in real time as new visual information becomes available. (A lightweight language model such as Phi-2 may optionally be used to convert rule outputs into concise natural language navigation instructions before text-to-speech generation.)

6.2. Dataset Sources

The perception components of the system will rely on publicly available datasets commonly used in computer vision research for indoor environments.

- **Object Detection**
  The obstacle detection module will use models trained on datasets such as the Microsoft COCO Dataset, which contains a large collection of annotated images across multiple object categories including common indoor obstacles such as chairs, tables, sofas, and people. These categories are relevant to indoor navigation scenarios where obstacles may obstruct the walking path.
- **Depth Estimation**
  Monocular depth estimation models used in the system will rely on pretrained models trained on indoor datasets such as NYU Depth v2, which provides RGB images paired with depth maps captured in indoor environments. This widely used dataset shall be leveraged for evaluating depth estimation algorithms in indoor scenes.

- **Custom Evaluation Data**
  In addition to publicly available datasets used for training or pretraining perception models, a small custom dataset of indoor recordings may be collected in controlled environments such as corridors and rooms containing furniture. These recordings will be used during evaluation experiments to simulate realistic obstacle navigation scenarios.

6.3. Hardware & Deployment Assumptions

The prototype system will use a monocular RGB camera positioned at approximately chest level and facing forward so that the captured image corresponds closely to the user's walking path.
Image processing and model inference will be executed on a computing device capable of running deep learning models in near real time, such as a laptop equipped with a GPU (or an edge AI device.)
The system is designed to operate using standard video resolutions commonly supported by consumer cameras. The primary objective is to generate navigation instructions quickly enough to allow users to respond safely while walking.

7. Testing and Evaluation Strategy

The effectiveness of the proposed action-oriented navigation system will be evaluated through **controlled indoor navigation experiments** designed to directly reflect the problem setting and risks outlined in the project statement.

7.1. Experimental Setup

Evaluation will be conducted in a **fixed indoor obstacle course** representative of common indoor environments such as homes or offices. The course will include static obstacles such as furniture and narrow passages that may obstruct the user's walking path within a near-field range of 0–3 meters.

Participants will be **blindfolded sighted individuals,** and each participant will navigate the same obstacle course under two system configurations:

- **Baseline system:**
  Object detection and depth estimation with audio feedback limited to object category announcements.
- **Proposed system:**
  Object detection and depth estimation combined with rule-based reasoning that generates explicit movement instructions such as *"move left"*, *"move right"*, or *"stop"*.

Each participant will perform multiple repeated trials under both system modes.

7.2. Evaluation Metrics

Performance will be assessed using metrics that directly measure **navigation safety, continuity, and responsiveness**, rather than perception accuracy alone. The following metrics will be recorded:

- **Collision Count:**
  The number of physical contacts between the participant and obstacles. A collision is defined as any physical contact occurring during navigation.
- **Navigation Completion Time:**
  The total time taken by the participant to complete the obstacle course.
- **Corrective Stops:**
  The number of instances where the participant performs a full halt (≥ 3 seconds), steps back, or requires manual intervention to continue.
- **Reaction Time:**
  The time elapsed between the delivery of an audio instruction and the participant initiating a corresponding movement response.

7.2.1. Success Criteria

To ensure that the evaluation produces clear and objective conclusions, quantitative success criteria are defined relative to baseline performance reported in prior research on assistive navigation systems.

Existing studies evaluating navigation assistance systems for visually impaired users commonly measure performance using metrics such as navigation completion time, collision frequency, and perception accuracy.
Neugebauer et al. (2020) evaluated a visual-to-auditory sensory substitution navigation system in indoor obstacle-course experiments with blind participants. Their results show that when using the sensory substitution device alone, participants required an average of 68.4 ± 54.7 seconds to complete the navigation task and experienced 2.96 ± 4.33 obstacle collisions per trial. In comparison, navigation using a white cane resulted in 27.0 ± 12.2 seconds completion time with 0.78 ± 1.28 collisions.

Recent computer-vision-based obstacle detection systems for visually impaired navigation also report perception performance metrics. Wang et al. (2024) evaluated several real-time object detection models for obstacle detection tasks and reported detection accuracies ranging from 21.22 mAP to 28.51 mAP, with small-object detection performance ranging from 8.52 mAPs to 18.93 mAPs depending on the detector architecture.
These values provide a reference point for evaluating the proposed system.

7.2.2. Baseline Values from Literature

Metric	Literature Baseline
Navigation completion time	~68 seconds (Neugebauer et al., 2020)
Collision count	~2.96 collisions per trial (Neugebauer et al., 2020)
Object detection accuracy	~8 - 19 mAPs (Wang et al., 2024)
Small-obstacle detection	~8-19 mAPs (Wang et al., 2024)

7.2.3. Target Improvement

The proposed system aims to improve navigation safety and usability by converting perception outputs into explicit movement guidance.
The system will be considered successful if it demonstrates the following improvements relative to baseline systems:

Metric	Literature Baseline
Navigation completion time	Comparable or improved relative to baseline (~68s)
Collision count	>= 30-50% reduction compared to perception-only baseline
Corrective stops	>= 30% reduction
Reaction time	<= 1 second user response delay

These criteria focus on improving navigation safety while maintaining continuous walking behaviour. The key expected improvement is a reduction in obstacle collisions and hesitation events through action-oriented navigation guidance.

(Since existing works primarily report perception accuracy rather than navigation performance, the evaluation focuses on user-centric metrics such as collision frequency and navigation continuity.)

7.3. Latency and Usability Evaluation

Since system response time cannot be predetermined prior to implementation, latency will be treated as a usability-driven requirement and measured during evaluation.

System latency will be defined as the time interval between:

- An obstacle entering the predefined walking zone in the camera frame, and
- The delivery of the corresponding audio instruction to the user.

Latency measurements will be analysed in sync with navigation performance to determine whether system response time supports safe and uninterrupted walking.

The system will be considered acceptable if it satisfies the following criteria:

- Navigation guidance allows the participant to react without stopping abruptly.
- The participant is able to walk continuously without frequent stops caused by delayed feedback.

7.4. Evaluation Objective

The objective of this evaluation is to quantify improvements in safe indoor mobility when visual perception is converted into actionable movement guidance rather than simple object identification. By directly comparing the proposed system against a perception-only baseline under identical conditions, the evaluation isolates the impact of decision-level reasoning on real-world navigation performance.

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